

A Hurwitz counterexample to cyclic Kaplansky approximants

Candidate answer to Question 9.1 of arXiv:2105.01110

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Status. Candidate counterexample; likely valid, pending expert review. Question 9.1 of Aleman–Hartz–McCarthy–Richter [1] has a negative answer as literally stated, already for the Hardy space on the disk.

1 Source question

For a normalized complete Pick space $\mathcal{H}$, write $[m]$ for the $\mathcal{H}$ -closure of $\text{Mult}(\mathcal{H})m$. Lemma 3.3 of the source says that if

$$m \in \text{Mult}(\mathcal{H}), \quad \varphi \in \text{Mult}(\mathcal{H}) \cap [m], \quad \|\varphi\|_{\text{Mult}(\mathcal{H})} \leq 1,$$

then there are $\varphi_n \in \text{Mult}(\mathcal{H})$ such that

$$\varphi_n m \longrightarrow \varphi \quad \text{weak-}^*, \quad \|\varphi_n m\|_{\text{Mult}(\mathcal{H})} \leq 1.$$

Question 9.1 asks whether the φ_n can always be chosen cyclic [1, Question 9.1].

If $\mathcal{V}$ is a locally convex vector space with dual $\mathcal{V}^*$, there is a finest topology on $\mathcal{V}$ with this dual, called the Mackey topology for the pair. Certain conditions on a topology force it to coincide with the Mackey topology; these include being metrizable (as d_F is) or being barrelled (as $\mathcal{I}$ is) [28, 10-1.9]. Putting these facts together, we get the following conclusion.

Theorem 8.6. *On $N^+(H_d^2)$, the inductive limit topology $\mathcal{I}$ and the metric topology d_F coincide.*

For the convenience of the reader, here is a direct proof.

Proof. As the by Theorem 1.23 and Proposition 8.5 identity map is continuous from $(N^+(H_d^2), \mathcal{I})$ to $(N^+(H_d^2), d_F)$ by the discussion preceding the theorem, we need to prove that the identity map from $(N^+(H_d^2), d_F)$ to $(N^+(H_d^2), \mathcal{I})$ is also continuous. To do this, it suffices to show that for each continuous semi-norm p on $(N^+(H_d^2), \mathcal{I})$, the inclusion (N^+, d_F) to $(N^+(H_d^2), p)$ is continuous, since the topology on any locally convex space is give by its continuous semi-norms (see e.g. [11, 1.10.1]).

So let (f_n) be a sequence in $(N^+(H_d^2), d_F)$ tending to zero. We claim that (f_n) tends to zero weakly in $(N^+(H_d^2), p)$. Indeed, if Γ is a continuous linear functional on $(N^+(H_d^2), p)$, then Γ is continuous on $(N^+(H_d^2), \mathcal{I})$ and hence on $(N^+(H_d^2), d_F)$ by Theorem 1.23 and Proposition 8.5. Thus, $\Gamma(f_n)$ tends to 0 for all Γ , establishing weak convergence. The uniform boundedness principle for the semi-normed space $(N^+(H_d^2), p)$ implies that $\sup_n p(f_n) < \infty$. (This follows from the uniform boundedness principle for normed spaces by taking a quotient.) So we have shown that if $d_F(f_n, 0) \rightarrow 0$, then $p(f_n)$ is bounded. But if $d_F(f_n, 0) \rightarrow 0$, then by passing to a subsequence, we may achieve that $d_F(nf_n, 0) \rightarrow 0$, hence $p(nf_n)$ is bounded, so (f_n) tends to 0 in $(N^+(H_d^2), p)$. This proves continuity of $(N^+(H_d^2), d_F) \rightarrow (N^+(H_d^2), p)$. $\square$

9. OPEN QUESTIONS

Lemma 6.4 says that if m is the lift of an outer function, then one can find cyclic multipliers φ_n so that $\varphi_n m$ is in the unit ball and tends to 1 weak-*. Is this true for all cyclic m ?

Question 9.1. *Can Lemma 3.3 be improved to conclude that each φ_n can be chosen to be cyclic?*

As pointed out in Section 1, when $d = 1$ there is no difference between the common range of co-analytic Toeplitz operators with cyclic symbols, and all non-zero co-analytic Toeplitz operators. Is this true for $d \geq 2$?

Question 9.2. *What is $\bigcap \{\text{ran}(T_m^*) : m \in \text{Mult}(H_d^2), m \neq 0\}$?*

In [19], M. Nawrocki showed that the dual of the full Smirnov class on the ball $\mathbb{B}_d$ consists of those functionals Γ satisfying

$$(9.3) \quad |\Gamma(z^\alpha)| \leq M \omega_\alpha e^{-c|\alpha|^{d/(d+1)}}.$$

2 Proof intuition

Set $m = 1$. The proposed strengthening would then make every contractive multiplier a bounded weak-* limit of cyclic multipliers. But cyclic multipliers cannot vanish: all their multiplier multiples share any interior zero, so they could not approximate the constant function 1 in the Hilbert space. Bounded weak-* convergence of multipliers implies locally uniform convergence. The target $\varphi(z) = z$ has a zero without being identically zero, and Hurwitz's theorem says it cannot be a locally uniform limit of zero-free holomorphic functions.

3 Counterexample

Theorem 1. *The improvement proposed in Question 9.1 is false. One may take*

$$\mathcal{H} = H^2(\mathbb{D}), \quad m = 1, \quad \varphi(z) = z.$$

Proof. The Hardy space $H^2(\mathbb{D})$ is a normalized complete Pick space, with kernel $(1 - z\bar{w})^{-1}$, and its multiplier algebra is $H^\infty(\mathbb{D})$. The multiplier $m = 1$ is cyclic. Moreover, $\varphi(z) = z$ belongs to $H^\infty \cap [1]$ (indeed, it is itself a multiplier multiple of 1), and $\|\varphi\|_\infty = 1$. Thus all hypotheses of Lemma 3.3 are satisfied.

Suppose its conclusion could be strengthened as in Question 9.1. There would be cyclic multipliers $\varphi_n \in H^\infty$ with

$$\|\varphi_n\|_\infty \leq 1, \quad \varphi_n \longrightarrow z \text{ weak-* in } H^\infty. \quad (1)$$

Each φ_n is zero-free on $\mathbb{D}$. Indeed, if $\varphi_n(a) = 0$ for some $a \in \mathbb{D}$, then every element of $H^\infty \varphi_n$ vanishes at a . Point evaluation is continuous on H^2 , so the H^2 -closure $[\varphi_n]$ also consists of functions vanishing at a and cannot contain 1. This contradicts cyclicity.

On bounded subsets of the multiplier algebra, weak-* convergence implies pointwise convergence; this fact is also recalled immediately before Lemma 3.3 in the source [1]. The uniform bound in (1) and Montel's theorem then upgrade the convergence to local uniform convergence. For completeness, every subsequence has a locally uniformly convergent subsequence, and its pointwise limit must be z ; this forces the whole sequence to converge locally uniformly to z .

Hurwitz's theorem [2, Theorem 10.28] now says that the locally uniform limit of the zero-free holomorphic functions φ_n is either zero-free or identically zero. The function z is neither: it vanishes at 0 and is not the zero function. This contradiction proves the claim. $\square$

Stronger obstruction. The same proof shows that, with $m = 1$, no nonzero contractive multiplier having an interior zero can be the limit required by the proposed cyclic strengthening.

4 Verification, limitations, and novelty

- Every hypothesis is literal: $H^2(\mathbb{D})$ is a normalized complete Pick space; $m = 1$ and $\varphi = z$ are multipliers; m is cyclic; $\varphi \in [m]$; and $\|\varphi\| = 1$.
- No identification of weak-* and local-uniform topologies is assumed without boundedness. The norm bound is part of Lemma 3.3, and Montel is used only after that bound.
- The counterexample answers the formal Question 9.1, which asks to strengthen Lemma 3.3 for arbitrary admissible φ . It does *not* settle the narrower preceding prose question: for an arbitrary cyclic m , can cyclic φ_n be found with $\varphi_n m \rightarrow 1$? Our target is $\varphi = z$, not 1.

- The parsed source is revised 21 April 2026 and still states Question 9.1. Bounded run-index and web/arXiv searches on 13 August 2026 included the exact question, “cyclic Kaplansky”, weak-* cyclicity, and later papers citing Lemma 3.3. No explicit answer or this counterexample was found. The argument is elementary, so novelty confidence is moderate rather than high.

Human-review recommendation. Verify that the intended quantifiers in Question 9.1 are those of Lemma 3.3, not only the special target $\varphi = 1$ mentioned in the preceding prose. Under the literal wording, the mathematical counterexample is complete.

References

- [1] A. Aleman, M. Hartz, J. E. McCarthy, and S. Richter, *The common range of co-analytic Toeplitz operators on the Drury–Arveson space*, Journal d’Analyse Mathématique 150 (2023), 215–247; arXiv:2105.01110, revised 2026.
- [2] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.